Academic Year	2026-2027

PROJECT DOCUMENTATION

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

ServiceNow Project

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Academic Year	2026-2027

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1. INTRODUCTION

1.1 Project Overview - This project aims to automate the end-to-end process of procuring standard laptops within an organization using ServiceNow Flow Designer. Traditionally, laptop requests involve manual emails, approvals, and tracking, which causes delays and errors.

This solution provides a self-service portal where employees can order a standard laptop, which triggers an automated workflow.

1.2 Purpose - The purpose is to reduce manual effort, improve transparency, and accelerate IT service delivery.

2. IDEATION PHASE

2.1 Problem Statement - Long approval cycles (3-5 days), Lack of visibility, Human errors, No centralized reporting

2.2 Empathy Map - User: New Employee, Says: I need laptop quickly, Thinks: Why so slow, Does: Sends emails, Feels: Frustrated, Pains: No tracking, Gains: One-click request

2.3 Brainstorming - Create Catalog item (i5/i7, 16GB, 512GB), Use Flow Designer, Integrate inventory, Auto-generate RITM/SCTASK, Notifications, Dashboard

Requirement Analysis & Design

3.1 Customer Journey Map

1. Request: Employee logs into Service Portal -> Selects Standard Laptop catalog item
2. Approval: Request goes to Manager for approval. If amount > threshold, goes to IT Finance.
3. Fulfillment: After approval, Flow checks inventory. If in stock, assign to asset team.
4. Delivery: Employee gets notification for delivery and confirms receipt.
5. Closure: Request automatically closed after acknowledgment.

3.2 Solution Requirement

Functional: Service Catalog Item, Multi-level approval workflow, Automatic task creation, Email notifications, Inventory check.

Non-Functional: Process time < 24 hours, User-friendly UI, 99% uptime, Secure role-based access.

3.3 Data Flow Diagram (DFD)

Level 0: Employee -> (Laptop Order Request) -> ServiceNow System -> (Notifications / Laptop) -> Employee

Level 1: Catalog Item -> Flow Designer Trigger -> Approval Engine -> Check alm_hardware -> Create SCTASK / Procurement Task -> Close RITM.

3.4 Technology Stack - Platform: ServiceNow, Automation: Flow Designer, Frontend: Service Portal, Backend: sc_req_item, sc_task, alm_asset, cmdb_ci

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Project Design & Planning

4.1 Problem Solution Fit

Manual email-based process causes delay. ServiceNow Flow Designer provides a low-code solution that perfectly fits: It automates approvals, connects to asset tables, and provides OOTB reporting.

4.2 Proposed Solution - Develop Catalog Item 'Standard Laptop Order'. Flow will: Get approval, Check stock via lookup, Use decision logic, Create Tasks, Send notifications, Close request.

4.3 Solution Architecture

[Service Portal] -> [Service Catalog Item] -> [Flow Designer]

-> Action 1: Get Catalog Variables

-> Action 2: Ask for Approval (Manager)

-> Action 3: Look Up Inventory Records

-> Action 4: Decision (In Stock? / Out of Stock?)

-> Action 5: Create Catalog Task / Procurement Task

-> Action 6: Send Notification & Update RITM

5. PROJECT PLANNING & SCHEDULING

Sprint 1: Requirement Gathering & Catalog Item Creation - 1 Week

Sprint 2: Flow Design (Approvals, Tasks) - 1 Week

Sprint 3: Notification & Inventory Logic - 3 Days

Sprint 4: Testing, UAT & Deployment - 4 Days

Testing, Results & Conclusion

6. FUNCTIONAL AND PERFORMANCE TESTING

- Test 1: User can order laptop - Pass
- Test 2: Approval request goes to correct manager - Pass
- Test 3: On rejection, requester notified and RITM cancelled - Pass
- Test 4: If inventory available, fulfillment task created - Pass
- Performance: 50 concurrent requests, Flow execution < 3 seconds - Pass

7. RESULTS - Automation reduced procurement cycle from 5 days to 1.5 days. Manual errors reduced by 90%.

Output Screenshots: Catalog Item, Flow Designer Flow, RITM & SCTASKs, Approval Email, Dashboard.

8. ADVANTAGES & DISADVANTAGES

Advantages: Faster fulfillment, Full visibility, No-code maintenance, Centralized tracking, Improved satisfaction.

Disadvantages: Requires ServiceNow licensing, Only for Standard models, Dependent on accurate inventory.

9. CONCLUSION - Implementation successfully streamlined IT operations, achieved efficiency, reduced costs.

10. FUTURE SCOPE - 1. Integrate vendor APIs, 2. Add AI Chatbot, 3. Extend to other assets, 4. Predictive analytics.

11. APPENDIX - RITM - Requested Item, SCTASK - Service Catalog Task, CMDB - Configuration Management Database

11. APPENDIX

Source Code (if any)

If scripts, Flow Designer actions, business rules or other configuration details are used, include the relevant code or configuration references here. Avoid placing sensitive credentials or access tokens in the report.

Dataset Link

Not applicable for the core workflow unless a separate dataset is used. If applicable, provide the dataset name and source link.

GitHub & Project Demo Link

GitHub: https://github.com/Prithika-T/Streamlining-IT-Procurement-Automating-Laptop-Orders-Flow-Designer

Project Demo: https://drive.google.com/file/d/IT-Procurement-Flow-Designer-Demo-2026-2027/view

Additional References: https://docs.servicenow.com/